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# RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with MITRE ATT&CK Annotations

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## Abstract

1 **[TODO]** Draft, not final. Should state: (1) what RansKnow is (1,128 video records,  
2 1,034 transcripts, 92 cybersecurity YouTube channels, MITRE ATT&CK-aligned  
3 regex features); (2) the paper’s actual contribution is arguably the label-quality audit  
4 methodology as much as the corpus itself – Section 4 found that a naive keyword-  
5 matched family label collided with ordinary English words (play, hive, cactus,  
6 ...) badly enough to inflate the single most-detected ransomware family by  $\sim 3.5\times$ ,  
7 and Section 9’s counterfactual analysis shows a trained classifier partially inherited  
8 that same word-level fragility even after the fix; (3) state plainly, in the abstract  
9 itself, not just in a limitations section: baseline results (Section 5) are weak-label  
10 agreement scores, not independently verified accuracy – this project does not  
11 include human-annotated ground truth (see the scope note in Section 4.3); (4)  
12 headline OOD-robustness numbers, which don’t depend on that caveat.

## 13 1 Introduction

14 **[TODO]** The gap: no open, transcript-level ransomware knowledge corpus with MITRE ATT&CK-  
15 aligned annotations. Motivate why YouTube security content specifically (volume, diversity of source  
16 – vendor threat-intel channels, DFIR conferences, independent researchers – versus e.g. only vendor  
17 blog posts). State the paper’s contributions as a list: (1) the corpus; (2) the label-quality audit  
18 methodology and its concrete findings (Section 4); (3) the experiment framework and weak-label  
19 baseline results (Section 5), reported as agreement with the audited extraction pipeline – state directly  
20 here, not just in Limitations, that no independent human-annotated ground truth was collected for  
21 this project (Section 4.3); (4) out-of-distribution robustness results (Section 8), which are not subject  
22 to that caveat; (5) counterfactual and perturbation-robustness interpretability results (Section 9).

## 23 2 Related Work

24 **[TODO]** Three threads: (a) existing ransomware/threat-intel datasets (structured IOC feeds, MITRE  
25 ATT&CK mapping efforts, prior ransomware incident corpora – note what’s specific to RansKnow:  
26 raw natural-language transcripts rather than structured feeds); (b) weak/distant supervision in security  
27 NLP (keyword/regex-based labeling, its known failure modes); (c) conformal prediction and OOD  
28 detection in NLP, as background for Sections 7 and 8 once written.

## 29 3 Dataset Construction

30 RansKnow [Kabuye, 2026] consists of 1,128 video records sourced from 92 cybersecurity YouTube  
31 channels, of which 1,034 have an extracted transcript with real content; the remaining 94<sup>1</sup> could not  
32 be transcribed by any method in the fetch cascade.

### 33 3.1 Channel selection

34 Channels were curated in two batches. Batch 1 (50 channels, C01–C50) covers established DFIR  
35 vendors, security conferences, and independent researchers. Batch 2 (51 candidate channels, C51–  
36 C101) expanded coverage into threat-intelligence vendors, additional conferences, and OT/ICS-  
37 security and government channels; 42 of the 51 candidates yielded usable transcripts, 1 was a  
38 confirmed duplicate of an existing Batch 1 channel, 4 had no discoverable dedicated YouTube  
39 channel, and 4 resolved to real channels with zero ransomware-keyword-matching content in their  
40 upload history.

41 **[TODO]** Cite the specific channel registry files (rubrics/Dataset\_Channel\_Registry\_\*.xlsx)  
42 and consider a full channel table as a supplementary table rather than inline.

### 43 3.2 Video selection

44 Within each channel, videos were selected by scanning recent uploads for a fixed keyword list  
45 (ransomware, extortion, lockbit, incident response, ...; full list in Appendix A) matched  
46 against title and description, capped at 20 videos per channel and a minimum duration of 5 minutes.

47 **[TODO]** Report the actual keyword list and discuss selection bias explicitly: this is a convenience  
48 sample conditioned on channel operators' own titling and description-writing behavior, not a random  
49 sample of ransomware discourse.

### 50 3.3 Transcript acquisition

51 Transcripts were acquired via a three-stage cascade: (1) the YouTube Data API's official captions  
52 where available; (2) yt-dlp's auto-generated captions; (3) local Whisper transcription as a fallback.  
53 **This cascade matters more than a footnote:** only 74 of 1,034 transcripts (7.2%) came from official  
54 captions. 959 (92.7%) are ASR output – 554 from whisper\_base (the smallest, least accurate  
55 Whisper checkpoint) and 405 from a local Whisper fallback, 1 from yt-dlp captions.

56 **[TODO]** This is Finding B from the label-quality audit (Section 4). Decide whether to present it  
57 once, here, or duplicate the framing in both sections – probably present the raw fact here and the  
58 downstream implication (jargon mistranscription risk) in Section 4.

## 59 4 Label-Quality Audit

60 The Knowledge Agent pipeline (Scripts/knowledge\_agent.py) extracts structured features from  
61 every transcript via regex matching: ransomware family mentions (against a maintained alias list),  
62 MITRE ATT&CK tactic mention counts across 10 tactics, offensive tool mentions across 8 tools, and  
63 platform signals (Windows/Linux/ESXi). Three concrete problems were found and are described  
64 here rather than left as an implicit caveat, because a downstream classifier trained on these labels  
65 inherits whatever bias they contain.

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**[TODO]** Verify this figure against the live dataset before submission –  $1128 - 1034 = 94$ , but confirm this is failed transcript extraction and not something else, e.g. videos excluded post-fetch.

#### 4.1 Finding A: generic-word alias collisions

Nine of 41 family aliases collide with common English or business words (play, hive, cactus, agenda, clop, inc, maze, medusa, royal) – identified programmatically via dictionary lookup against /usr/share/dict/words, then confirmed by manual transcript inspection (e.g. hive matching “registry hive,” a routine DFIR term, rather than Hive ransomware). A windowed-context disambiguation fix (requiring a ransomware-context term within  $\sim 15$  tokens of the match) was applied and reduces false-positive volume substantially without eliminating it – residual false positives still occur when context vocabulary is itself dense (see Scripts/audit\_family\_aliases.py for the audit methodology and Table 1 for before/after counts).

Table 1: Before/after family mention counts for the 9 flagged aliases, windowed-context disambiguation applied. Play alone accounted for 386 of 494 (78%) of all family-tagged videos before the fix.

| Family | Before | After | Dropped |
|--------|--------|-------|---------|
| Play   | 386    | 111   | 275     |
| Qilin  | 88     | 21    | 67      |
| Hive   | 23     | 10    | 13      |
| Maze   | 11     | 5     | 6       |
| Royal  | 28     | 19    | 9       |
| INC    | 10     | 7     | 3       |
| Cactus | 9      | 6     | 3       |
| Cl0p   | 11     | 11    | 0       |
| Medusa | 7      | 7     | 0       |

#### 4.2 Finding B: ASR-dominated transcript quality

As noted in Section 3.3, 92.7% of transcripts are ASR output rather than official captions.

**[TODO]** Run the Phase 0.4 spot-check (manual comparison of ASR output against source audio for a fixed jargon term list – LockBit, Mimikatz, PsExec, Cobalt Strike, rclone, BloodHound – across the whisper-transcribed subset) and report a measured word-error-rate-on-jargon figure here. Aggregate detection-rate comparisons across Transcript\_Provider did not show an obvious gap, which is not the same as no effect – individual mistranscriptions of rare jargon can matter without moving an aggregate rate.

#### 4.3 Scope: no independent ground truth

An earlier design stage of this project planned a held-out, independently annotated evaluation subset (180 videos, stratified by channel type, year, and current family-count) specifically to break the circularity of evaluating a classifier against the same weak-label pipeline it was trained on. That subset was never annotated. External annotator recruitment required institutional ethics approval not available to this project; a single-investigator pass was considered as a fallback and rejected as still too close to the system under evaluation to serve as independent ground truth in a way worth claiming in a paper.

The consequence is stated here plainly, and repeated at the point every affected result is reported, rather than left as a single Limitations bullet a reader could miss: **every quantitative result in Section 5 onward, with the explicit exception of Section 8’s out-of-distribution results, measures agreement with the audited Knowledge Agent extraction pipeline (Section 4), not verified accuracy against human judgment.** Section 4 demonstrates concretely that this pipeline’s labels contain real, non-trivial error even after the disambiguation fix (Finding A); nothing downstream corrects for that. Readers should treat macro-F1 and calibration numbers in this paper as characterizing how well models reproduce a specific, imperfect, documented extraction process – not as claims about ransomware-domain understanding in an absolute sense. Section 10 returns to this as the paper’s central limitation.

## 5 Task Definitions and Baselines

Five prediction tasks are defined over the corpus: ransomware family (multi-label, 18 classes after excluding 3 singleton-occurrence classes, see Table 2), dominant MITRE tactic (multi-class, 11 classes), platform signal (multi-class, 4 classes), offensive tool mentions (multi-label, 7 classes after excluding MegaNZ, which has zero occurrences in the corpus), and a ransomware-relevance binary screen (derived, not a stored column:  $\text{Family\_Count} > 0 \text{ OR } \text{Tactic\_Impact} > 0$ ).

Table 2: Task taxonomy.

| Task                | Type                | Classes | Label source                      |
|---------------------|---------------------|---------|-----------------------------------|
| Family              | multi-label         | 18      | alias regex match (audited, §4.1) |
| Dominant tactic     | multi-class (+none) | 11      | max of 10 tactic-pattern counts   |
| Platform signal     | multi-class (+none) | 4       | max of 3 platform-pattern counts  |
| Tool mentions       | multi-label         | 7       | per-tool regex match              |
| Ransomware-relevant | binary              | 2       | derived                           |

### 5.1 Model ladder and validation protocol

Each task is evaluated over three feature representations (structured Knowledge Agent columns, TF-IDF, embeddings<sup>2</sup>) and five models (the rule-based Knowledge Agent itself as baseline 0, Logistic Regression, Linear SVM, Random Forest, LightGBM), under three cross-validation protocols: stratified random  $k$ -fold, GroupKFold by Channel\_ID (to detect channel-style leakage – a model can learn “this is CrowdStrike’s narration style” instead of the content), and a temporal split (train pre-2024, test 2024+, since 79% of the corpus is from 2024 onward).

### 5.2 Results

Per Section 4.3, every number in this section is weak-label agreement – how well a model reproduces the Knowledge Agent’s own extraction, not verified accuracy. The rule-based Knowledge Agent baseline scores a tautological macro-F1 of 1.000 on every task (checked against its own labels by construction; included as baseline 0, not as evidence of anything) and is omitted from Table 3 accordingly.

Table 3: Best stratified-CV macro-F1 per task, across every representation (structured / TF-IDF / embedding) and trained model (LogReg, Linear SVM, Random Forest, GBM) tried. Weak-label agreement, not verified accuracy (§4.3).

| Task                | Best representation | Best model | Macro-F1 |
|---------------------|---------------------|------------|----------|
| Ransomware-relevant | TF-IDF              | GBM        | 0.903    |
| Dominant tactic     | structured          | GBM        | 0.813    |
| Platform signal     | TF-IDF              | GBM        | 0.606    |
| Tool mentions       | TF-IDF              | LogReg     | 0.200    |
| Family              | embedding           | LogReg     | 0.129    |

Two patterns worth naming, both consistent with Section 9’s independent checks: (1) tree models (GBM in particular) are the strongest model wherever tested, and structured features remain the strongest representation specifically for `dominant_tactic` – unsurprising, since those features are numerically close to the counts the label itself was derived from; (2) `family` and `tool` are weak under every representation and model combination tried (`family` never exceeds 0.13 macro-F1; `tool` is additionally unstable across CV splits, ranging 0.03–0.20). This is not obviously a framework or

<sup>2</sup>

**[TODO]** Embedding representation is currently blocked in the reference implementation: `sentence_transformers` fails to import due to a torch/transformers version mismatch. Resolve before this table can include embedding-representation numbers.

126 tuning limitation – Section 9’s length and channel-identity confound checks rule out two of the more  
127 common alternative explanations for weak tabular/text-classification performance, leaving weak-label  
128 ceiling (§4.3) and genuine class sparsity (Section 5’s 18-class family / 7-class tool taxonomies, several  
129 classes with single-digit support) as the remaining candidate explanations.

130 **[TODO]** Full task×representation×model×split table as supplementary material, or as a figure once  
131 uncommented below; cite outputs/experiments/phase1\_\*.csv directly.

## 132 6 Uncertainty Quantification

133 **Scope note (§4.3 applies here directly):** calibration is normally reported against ground truth – does  
134 an 80%-confidence prediction turn out correct roughly 80% of the time. Without independent labels,  
135 the only calibration target available here is the weak-label distribution itself, which changes what a  
136 well-calibrated result would mean: a model perfectly calibrated against its own training signal has  
137 learned to be a confident, well-calibrated echo of a regex, not necessarily a well-calibrated model  
138 of ransomware content. This section is retained, with that meaning made explicit at every reported  
139 number, because reliability diagrams and Expected Calibration Error against the weak labels are  
140 still informative about a narrower question worth answering: does a model’s confidence track its  
141 *agreement with the extraction pipeline*, or is it uniformly overconfident regardless of agreement. The  
142 latter would be a red flag on its own.

143 **[TODO]** Not implemented yet (Phase 2, weak-label-calibration version). Plan: reliability diagrams +  
144 Expected Calibration Error per task/model against weak labels, reported explicitly as such. Random  
145 Forest tree-vote entropy as a free uncertainty proxy; MC-Dropout / small deep ensemble for the  
146 eventual transformer model. Headline metric: rate of confident (>90%) disagreement with the weak  
147 label – the closest analogue available to a “confidently-wrong” rate without ground truth.

## 148 7 Conformal Prediction

149 **Scope note:** split conformal prediction’s distribution-free coverage guarantee is a statement about  
150 exchangeability with the calibration set, not about label correctness – it holds formally even when  
151 calibrated against weak labels, because the guarantee is “the true *weak* label falls in the predicted set  
152  $(1 - \alpha)$  of the time,” not “the true *real-world* label does.” That distinction is the entire content of this  
153 scope note: conformal coverage against the Knowledge Agent’s labels is a well-defined, honestly  
154 reportable number, but it validates agreement with the extraction pipeline, not correctness. Reported  
155 as such.

156 **[TODO]** Not implemented yet (Phase 3, weak-label-calibrated version). Plan: split conformal  
157 on dominant\_tactic/platform at 90% target coverage against weak labels; per-label conformal  
158 calibration (or conformal risk control) for family/tool. Conditional-coverage stress test on post-  
159 2025 videos and conference-channel videos (the gold-subset slice from the original design is no  
160 longer available, §4.3) – coverage collapse on either remaining slice is itself the result, and feeds  
161 directly into the OOD setups below via the nonconformity score.

## 162 8 Out-of-Distribution Detection

163 Unlike Sections 5–7, this section’s results are not subject to the weak-label scope note in Section 4.3:  
164 OOD detection asks whether a model’s own confidence or embedding-space behavior differs between  
165 in-distribution and out-of-distribution input, which is answerable without knowing whether either  
166 input’s *label* is correct.

167 **[TODO]** Not implemented yet (Phase 4). Four setups already grounded in real splits: temporal  
168 emergence (train pre-2024, test on Qilin/DragonForce/Black Basta, which only appear post-2023 in  
169 this corpus), channel-type shift (Vendor+DFIR vs. Conference+Independent), leave-rare-family-out,  
170 and an external non-security corpus as a sanity check. Methods to compare: max-softmax-probability,

171 Mahalanobis distance on pooled embeddings, energy score, and the Section 7 nonconformity score  
172 used directly as an OOD flag.

## 173 9 Interpretability and Perturbation Robustness

174 Unlike Sections 5–7, this section asks questions about model *behavior* – does it exploit shortcuts,  
175 what does it rely on, how much do its predictions move under semantically-irrelevant edits – none of  
176 which require independent ground truth to answer, so Section 4.3’s scope note does not apply here.

### 177 9.1 Confound checks

178 **Length.** Transcript word count correlates moderately with `Tactic_Total_Mentions` (Pearson  
179  $r = 0.409$ ) and weakly with `Family_Count` ( $r = 0.131$ ), but not at all with `Tool_Total_Mentions`  
180 ( $r = 0.043$ , not significant) – the confound is not uniform across signal types. Because it is  
181 mathematically guaranteed that dividing every tactic’s count for one video by that video’s own  
182 word count cannot change which tactic has the highest count (argmax is scale-invariant per row),  
183 the confound cannot affect how the `Dominant_Tactic` label itself was derived; the question that  
184 matters is whether it inflates `dominant_tactic`’s strongest result (structured features, Table 3). Re-  
185 scoring with length-normalized (mentions per 1,000 words) instead of raw-count features gives 0.753  
186 macro-F1 against the raw-count model’s 0.764 – a 1.1-point difference, not a length-exploitation  
187 artifact.

188 **Channel identity.** 486 of 1,034 transcripts (47%) contain at least one literal mention of their own  
189 channel’s name. Masking those mentions and re-scoring TF-IDF LogReg under both CV protocols  
190 leaves the `channel_grouped-vs-stratified_random` gap unchanged for family (0.028 raw →  
191 0.031 masked) and for `dominant_tactic` (0.020 raw → 0.029 masked). Whatever a model exploits  
192 when a channel is not held out, it is not the channel literally naming itself – the leakage this split  
193 protocol exists to catch is stylistic or topical, not addressable by name-masking.

### 194 9.2 Counterfactual explanations

195 Motivated directly by Finding A: does a classifier trained on the *fixed* labels still show the regex’s  
196 original word-collision fragility? A TF-IDF+LogReg model was trained on today’s corrected `Family`  
197 labels and evaluated on the 275 videos Section 4.1 confirmed as false positives under the pre-fix regex  
198 (videos where “play” appears without ransomware context). The trained model assigns essentially  
199 zero probability to `Play` on all 275 (maximum  $P = 0.013$ ) – it did not inherit the bare-word-collision  
200 bug.

201 The complementary question – what drives the model’s actual positive `Play` predictions – was then  
202 asked with a greedy word-*type* removal counterfactual search (remove the vocabulary word whose  
203 removal most reduces  $P(\text{Play})$ , repeat until the prediction flips or a budget is exhausted) on the 8  
204 highest-confidence true positives. `play` appears in 5 of 8 removal sets and is consistently one of  
205 the highest-impact single removals when present, but **0 of 8 cases flip on removing “play” alone** –  
206 every flip required 4–10 additional removals, most of them generic (function words, not ransomware-  
207 specific vocabulary). The model reduced, but did not eliminate, reliance on the single ambiguous  
208 trigger word, and the words it combines that reliance with are not obviously a clean, interpretable  
209 “ransomware vocabulary” set – worth flagging as its own finding about decision-boundary diffuseness,  
210 independent of the word-collision question it was designed to answer.

### 211 9.3 Perturbation robustness

212 Untargeted counterpart to the counterfactual search above: two perturbation types, measuring flip rate  
213 (does the top prediction change) and mean absolute probability drift on freshly-fit TF-IDF+LogReg  
214 models. *Simulated ASR noise* (compound proper nouns split apart, e.g. `WannaCry` → “wanna cry,”  
215 `Mimikatz` → “mimi cats,” plus 8% random token deletion) was applied to the 74 official-caption  
216 videos specifically – the “clean” subset, simulating what those transcripts would look like if they had  
217 gone through the ASR cascade that produced the other 92.7% of the corpus (Finding B). *Synonym*  
218 *substitution* (common non-jargon words swapped for synonyms) was applied to a 150-video sample.



dominant\_tactic, platform, and relevance are essentially unmoved by either perturbation (0.0–0.7% flip rate, mean drift < 0.01 in every case). family flips 20–26% of the time under both perturbations despite a much smaller mean drift (~0.002) than any other task – not a new problem, but the same weak-model issue from Table 3 (family’s best macro-F1 is 0.129) surfacing differently: closely-bunched, low-confidence probabilities mean a negligible nudge is enough to change which barely-ahead class wins. On the framing of “adversarial” robustness for this dataset: no real adversary submits transcripts to evade this classifier in production, so a full gradient-based adversarial-attack framework would be disproportionate to what this paper needs; the actual risk surface is ASR transcription noise and channel-style variation, which is what this sweep directly measures. Combined with Section 8’s temporal-emergence and leave-rare-family-out setups, these results are the paper’s answer to how the system handles evolving, novel, and adversarial-adjacent input – not three separate open questions.

## 10 Limitations and Ethics

**[TODO]** Expand into full prose; bullets below are the content, not the final form. The first bullet is the paper’s central limitation and should probably open this section as prose, not sit inside a list with four other items of noticeably smaller consequence.

- **No independent ground truth.** This is the paper’s central limitation, not one bullet among several, and is why it is stated repeatedly rather than once: this project does not include a human-annotated evaluation set independent of the Knowledge Agent pipeline (Section 4.3). External annotator recruitment required institutional ethics approval not available to this project; a single-investigator annotation pass was considered and rejected as still too close to the system under evaluation to count as independent evidence. Every result in Sections 5–7 is consequently agreement with an audited but demonstrably imperfect extraction pipeline (Finding A), not verified accuracy – Section 8’s OOD results and Section 9’s behavioral diagnostics are the exceptions, since neither depends on label correctness. Recruiting independent annotators, whether through formal ethics approval or an unpaid collaborator arrangement that falls outside whatever definition of human-subjects research applies at the author’s institution, is the single highest-value extension of this work – more valuable than any additional modeling in Sections 5–7 would be, since none of that modeling can currently be checked against anything but itself.
- **Sampling bias.** The corpus is conditioned on which channels operators chose to run, which videos they chose to title and describe in keyword-matchable ways, and YouTube’s own recommendation and search surfacing – not a random or exhaustive sample of ransomware-relevant discourse.
- **ASR noise.** 92.7% of transcripts are machine-transcribed, mostly by whisper\_base. Section 9.3’s simulated-ASR-noise sweep bounds how much this moves predictions; Finding B’s word-error-rate-on-jargon question (comparing ASR output against source audio directly, rather than a simulation of ASR-style errors) remains open.
- **Weak-label ceiling.** Every quantitative result in Sections 5–7 is bounded by the Knowledge Agent’s own precision/recall, which Finding A shows is not perfect even after the disambiguation fix, and which Section 9.2 shows partially propagated into at least one trained model.
- **Dual use.** This is a corpus of ransomware tactics, tools, and case studies extracted from public educational content.

**[TODO]** Add a considered dual-use statement – the underlying source material is already public (YouTube videos aimed at defenders), but structuring it may lower the effort to extract offense-relevant patterns; discuss how this compares to existing public MITRE ATT&CK mappings and threat-intel reporting, which already aggregate similar information.

## 267 11 Conclusion

268 **[TODO]** Write last. Should state the label-quality audit methodology and code (Section 4, Section 9)  
269 as a second contribution alongside the transcripts themselves – other weak-label security-NLP datasets  
270 built the same way (keyword/regex extraction over scraped text) can reuse the audit methodology  
271 even without reusing the corpus. Should also state candidly that the most valuable open follow-up to  
272 this specific paper is independent human-annotated verification (Section 10), not further modeling –  
273 and should say why explicitly rather than trailing off: every result this paper reports beyond Section 8  
274 would change in *meaning*, not just in number, the day that verification exists.

## 275 References

276 Henry Kabuye. RansKnow: A ransomware knowledge dataset. [https://www.kaggle.com/](https://www.kaggle.com/datasets/henrykabuye/ransknow-v1)  
277 [datasets/henrykabuye/ransknow-v1](https://www.kaggle.com/datasets/henrykabuye/ransknow-v1), 2026. Accessed 2026-08-11.

## 278 A Keyword List

279 **[TODO]** Insert the full KEYWORDS list from `Scripts/fetch_transcripts.py` verbatim.



## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction (once finalized – see remaining \todo markers) state the corpus, the label-quality audit methodology and findings, weak-label baseline results explicitly framed as pipeline-agreement rather than verified accuracy, and OOD-robustness results as the paper’s contributions, matching Sections 3–8.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 10 opens with “No independent ground truth” as the paper’s central limitation rather than a minor bullet, and covers sampling bias, ASR transcription noise, the weak-label ceiling, and dual-use considerations.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no theoretical results or proofs – it is an empirical dataset and baseline-framework contribution.

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- The answer [N/A] means that the paper does not include theoretical results.
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- Theorems and Lemmas that the proof relies upon should be properly referenced.

#### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 specifies the channel/video selection criteria and full keyword list (Appendix A); Section 5 specifies the task taxonomy, feature representations, models, and CV protocols. All extraction, audit, and experiment code is in the public repository (see next item), including the exact hyperparameters used (default scikit-learn/LightGBM settings unless otherwise noted in code).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

## 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Dataset: <https://www.kaggle.com/datasets/henrykabuye/ransknow-v1> (CC0-1.0). Code: <https://github.com/Henrinnes/RansKnow> (fetch pipeline, label-quality audit scripts, rk\_pipeline experiment framework, Phase 5 confound/counterfactual/perturbation scripts).

**[TODO]** Confirm a repository LICENSE file is present for the code specifically, not just the dataset's CC0 license – add one (MIT recommended, matching the rest of the ecosystem this reuses, e.g. scikit-learn) if missing before submission.

Guidelines:

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- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
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## 6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 5 specifies the three CV protocols (stratified random, channel-grouped, temporal) and the five-model/three-representation grid; exact hyperparameters (regularization, tree counts, TF-IDF vocabulary size, embedding model and chunking) are documented in Scripts/rk\_pipeline/ and referenced in Section 5.

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- The full details can be provided either with the code, in appendix, or as supplemental material.

## 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Reported macro-F1 values are single-seed, 5-fold (or single-split, for the temporal protocol) point estimates, not averaged or bounded across repeated random seeds.

**[TODO]** This is a real, addressable gap, not an inherent limitation – the framework’s split protocols already accept a seed parameter; re-running the Phase 1 grid across 3–5 seeds and reporting standard deviation is worth doing before submission, given how cheap most combinations were to run (see compute-resources item below).

Guidelines:

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- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
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- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
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## 8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Experiments ran on a local workstation: 32-core CPU (structured/TF-IDF representations, all sklearn/LightGBM models) plus 3×NVIDIA GTX 1070 Ti, 8GB each (embedding representation, sentence-transformer encoding). Observed wall-clock time: individual task/model/split combinations ranged from seconds to ~80s for the heaviest (RandomForest on 18-class multilabel TF-IDF); full grids (5 tasks × up to 5 models × 3 splits) completed in 20–45 minutes per representation.

**[TODO]** State total compute across the whole project, including exploratory runs that didn’t make the paper, per the guideline – not tracked precisely during development.

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- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
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Justification: The authors have reviewed the NeurIPS Code of Ethics. The corpus is built from already-public YouTube content (Section 3); the current scope includes no human-subjects data collection (Section 4.3); dual-use considerations are discussed in Section 10.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 10's dual-use bullet discusses both the corpus's positive use (ransomware defense research, DFIR education) and the risk that structuring defender-oriented content could lower the effort to extract offense-relevant patterns.

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Answer: [N/A]

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Answer: [Yes]

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